

Sprint 4 Documentation – GenAI Worker

Team: Sebastian and Felix

Repository: <https://github.com/BunnySupreme/SWEN3Project>

Sprint duration: 04.12.2025 – 11.12.2025

Estimated workload: 6 h / person / week

1. Sprint Goal (from assignment)

1. Extend docker-compose to include configuration for GenAI service (paperless-geminisummarizer)
2. Add a new GenAI Worker service (GeminiSummarizer)
3. On document upload, after OCR is complete:
 - a. Send the extracted text to a GenAI API (Google Gemini)
 - b. Receive a summary as a response
4. Use logging in critical positions
5. Handle exceptions and API failures properly

2. Implemented Work

GenAI Worker (GeminiSummarizer)

1. Requests summaries from Google Gemini using an API Key
2. Sets prompt and parameters given to Google Gemini
3. Handles API failures and throws related exceptions
4. Returns summary
5. Provides route to request summary via HTTP

OCR Worker refactoring

1. A GeminiSummarizerClient class was added
 - a. Facilitates requests to GeminiSummarizer via HTTP
 - b. Catches and handles exceptions thrown by GeminiSummarizer
 - c. Uses interface IGeminiSummarizer to facilitate dependency inversion
2. OcrWorkerService was updated
 - a. GeminiSummarizerClient was injected (using the interface and AddHttpClient in the building stage in Program.cs)
 - b. Now calls GeminiSummarizerClient for summary production, previous placeholder code was removed

REST Server

1. Updating PostgreSQL DB with summary was already implemented, thus no changes had to be made to REST server

Adminer

1. Adminer was added to docker-compose to allow GUI for interaction with the PostgreSQL DB (CLI was used up to this point)

6. Workload (Estimate)

Team Member	Week / Sprint	Estimated Hours	Main Tasks
Sebastian	Sprint 5	6	OcrWorker, Adminer
Felix	Sprint 5	6	GeminiSummarizer